[Total No. of Question -04] [Total No. of Pages - 01]

# G H Raisoni International Skill Tech University, Pune

**F.Y B.Tech (TERM-II)**

**CAE I- Summer 2026 (2024Pattern)**

**ENGINEERING PHYSICS (25UBSL1204))**

**(Time: - 01 Hour)** **(Max. Marks: 20)**

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**Instructions to the students:**

***i) All questions are compulsory. ii) Neat diagrams must be drawn wherever necessary. iii) Figures to the right indicate full marks. iv) Use of scientific calculator is allowed v) Assume suitable data, if necessary.***

| Q.No | *Question* | Marks | CO | BL |
|---|---|---|---|---|
| Q1.(a) | Explain the concept of interference of light and state the conditions for sustained interference. | [5] | 1 | 2 |
| b) | Describe Young's double slit experiment with a neat diagram and explain the formation of bright and dark fringes. | [5] | 1 | 2 |
| | **OR** | | | |
| Q 2.(a) | Describe thin film interference due to reflected light and state the conditions for maxima and minima. | [5] | 1 | 2 |
| (b) | Explain the principle of anti-reflection coating and state its applications. | [5] | 1 | 2 |
| Q 3.(a) | Classify diffraction into Fresnel and Fraunhofer diffraction and explain any three differences between them. | [5] | 2 | 2 |
| (b) | Analyze the Fraunhofer diffraction pattern of a single slit and explain why the central maximum is brightest and widest. | [5] | 2 | 3 |
| | **OR** | | | |
| Q. 4 (a) | Describe the production of plane polarized light by reflection and state Brewster's law. | [5] | 2 | 2 |
| (b) | Evaluate the working of a polarizer–analyzer arrangement and analyze the variation of transmitted intensity with angle. | [5] | 2 | 3 |